

Week 4)

FRAMEWORKS + MODELS for EVALUATION

→ Developing and evaluating complex interventions
guidance on development of complex intervention
evaluation
implementation

key functions of evaluation stage
assessing effectiveness
understanding change process
assessing cost-effectiveness

Part 1 ← framework
key messages for each stage

choose appropriate study for evaluation.

Part 2 → further?

Part 3 → Case Studies

Robust trials → TAsMINH4

Generate

reliable

statistically significant results

ensure appropriate interpretation of results

Strength of statistical tests and analyses — unaffected by outliers
estimated effects of interventions not being sensitive to changes
levels of statistical significance
confidence interval
p value
in model specification

Determinants of robustness

summary statistic (relative risk reduction)

sample size
N of events

N of pts lost to follow up

TASMIN4

↑ sample size

describes sample size calculation

inclusion criteria

trial profile

complete + partial withdrawals

lost to follow up

lifetime progression

adverse events

Results for each outcome — reported with 95% confidence interval and p value

↓
≠ between 2 intervention group for each outcome and ≠ effect in subgroups and adverse events were also examined

Sensitivity analyses — conducted

Detailed statistical analyses prepared before final analysis and unmasking of treatment allocation

Process and engagement measures

→ Pervasive system design (article)

Define what the DH is trying to change

| to understand the

Oinas - Kukkonen

2009

design an intervention that is process

effective + promotes user

engagement + inform

what should be targeted + strategy

| to achieve intended outcomes

framework for

reinforce

shape attitudes / behaviours

change ~~attitudes~~

Developing + evaluating pervasive systems

3 phases:

1) Understanding key issues + 7 goals that need to be achieved

2) Analysing situation context

intent
event
strategy

} these help understand:
- nature of info technology

Design Insights

4 categories

translate insights from 2 previous phases into features + functionalities in the final intervention

- users
- their needs
- what they favourably respond to
- how pervasion works
- key players in pervasion process
- pervasion strategy
- what a system should embody

1) Primary task support → support carrying out of the users' primary task

2) Dialogue support → provision of feedback to help user continue to progress towards their goal / target behaviour

3) Credibility

4) Social support → leveraging social influence to motivate users

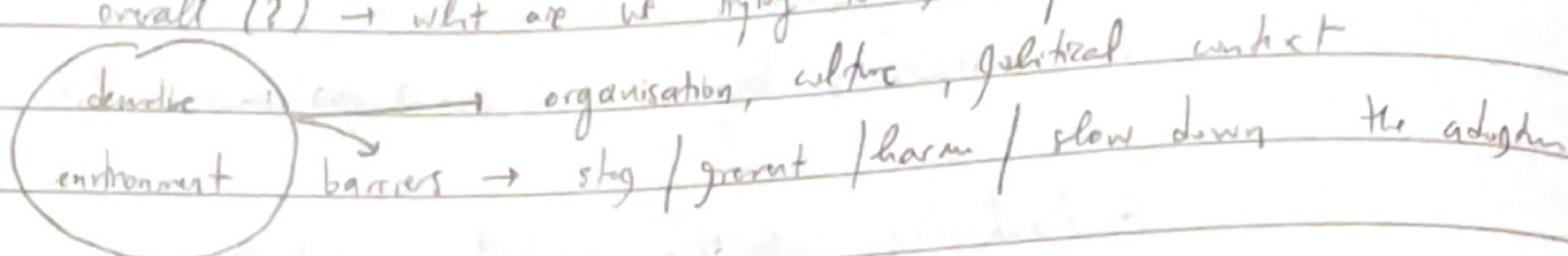
Success Models

LOGIC Models

→ Before intervention

→ Targeted

overall (?) → what are we trying to move / measure?



Structure / Process / Outcome

intervention	things we	long term	} things we can measure
money	want to	short term	
people	change		
resources			
	activities		

→ Digital health tends to be under-evaluated

health outcomes / behaviours I want to change

↓
INDICATORS

↓
experimental design / observational (?)

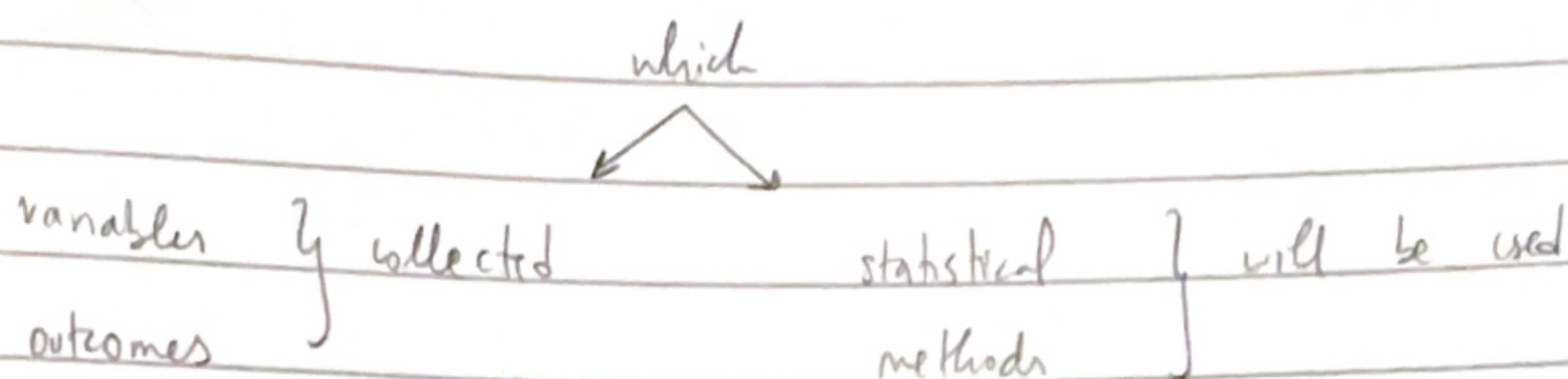
build + design my THEORY OF CHANGE

→ Think of unintended consequences → DARK LOGIC MODEL

STATISTICAL ANALYSIS PLAN

PACE - Walwyn, 2013

SAP → detailed technical elaboration of planned analysis



Benefits of SAP:

- Guides analysis + reporting
- demonstrate that findings → are faithful to the research protocol
- enables clear communication w/ statistician
- ↑ efficiency of data analysis
- deters, ↓ bias (especially selection / publication)
- (+) transparency
- suggests efforts to reproduce study results

⇒ Before data analysis

⇒ But graduate is to publish it

key elements =

Objectives + hypothesis

Study design

Primary + Secondary outcome measures

Sample size

Analysis method